Fiscal Year:	FY 2013	**Task Last Updated:**	FY 06/06/2013
PI Name:	Fike, John R. Ph.D.		
Project Title:	Use of a molecular marker of learning and memory to assess effects of 56Fe irradiation on hippocampus-dependent cognition and neurogenesis		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94110	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Space Radiobiology NNJ08ZSA001N
Start Date:	10/01/2008	**End Date:**	06/30/2013
No. of Post Docs:	2	**No. of PhD Degrees:**	2
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	4
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 6/30/2013 per PI and UCSF Admin. (Ed., 5/7/2013) NOTE: Extended to 9/30/2013 per PI and NSSC information (Ed., 8/2/2012)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Raber, Jacob (Oregon Health & Science University) Rosi, Susanna (University of California San Francisco)		
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Task Description:	Particulate irradiation in the space environment may pose a significant risk to space flight crews during a mission or upon return to Earth. This may involve the central nervous system, where low radiation doses can induce cognitive or behavioral impairments. Understanding the cellular and molecular basis of cognitive injury will facilitate the establishment of risk estimates and may provide a foundation for the establishment of effective countermeasures or treatments for such injury. Using an animal model, we will determine the extent to which low doses of 56Fe irradiation affect cognitive function and will determine if such changes are associated with the alteration of a gene that is associated with learning and memory, or with the production and maturation of newly born neurons.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The chosen animal model and the behavioral, cellular, and molecular endpoints to be used will provide new insight regarding how high LET irradiation affects factors associated with cognitive or behavioral function. Further, the design of these studies, using relatively low doses such as those that could be experienced in a long term space mission, will allow us to determine if there is a threshold of such effects. This could have significant impact in the determination of risk for astronauts during a prolonged mission. Additionally, understanding the association between neurogenesis, the expression of the immediate early gene Arc, and behavioral performance, will provide valuable new information about the how cognitive function is impacting by deleterious insults such as irradiation. Such information could have an impact in a number of biomedical fields including radiation oncology, neurodegeneration, and cognitive neuroscience.

SUPPLEMENTAL INFORMATION FOR FINAL REPORTING (JUNE 2013)
The overall goals of this project were to determine: (1) how low doses of 56Fe ions affected behavioral performance; (2) if specific cellular (e.g. neurogenesis) and molecular factors (immediate early gene Arc) might play a role in cognition; and (3) if those endpoints were affected by inflammatory changes. The experimental methods chosen to accomplish these goals provided a means by which the effects of low radiation doses could be quantified both in terms of behavioral performance as well as specific cellular and molecular endpoints that are associated with learning and memory.

Our proposed measures of behavioral performance, which were also used to induce Arc expression, involved a fear conditioning paradigm with 2 versions: one that required the hippocampus (context version) and one that did not require the hippocampus (cued version). The version that required the hippocampus consisted of two types of behavioral paradigms. The first involved no tone shock pairing (-/-) and represented the ability of an animal to recognize an environment previously encountered. Animals that recognized that environment would explore it less, a behavior that is referred to as spatial habituation learning. The second paradigm involved tone shock pairing (+/+) and represented a Pavlovian type of response; that is, animals received a mild shock the first day, and when subsequently placed in the same environment, they would freeze more if they remembered having received an adverse stimulus in that environment the day before. For each version used, thirty minutes after the end of the final testing session on day 2, tissues were collected and prepared for microscopic analyses.

In the current funding period we completed all analyses related to behavior, neurogenesis, Arc protein expression, and measures of the endogenous inflammatory cells in the brain, the microglia. These endpoints were assessed after whole body irradiation with either 0, 0.25 Gy, 0.5 Gy or 1.0 Gy of 56Fe. All animals underwent cognitive testing 3 months after irradiation using a fear conditioning training- and testing-paradigm. The three-month time point was selected because that time has been previously used to show cognitive deficits after photon or HZE irradiation, as well as changes in Arc expression. During the current funding period we quantified the effects of whole body 56Fe irradiation (600 MeV, 0.5 or 1 Gy) on hippocampus-dependent and hippocampus-independent cognitive performance and determined if those effects were associated with changes in Arc expression in the enclosed and free blades of the dentate gyrus. Whole body 56Fe irradiation impacted contextual but not cued fear freezing and the percentage of Arc-positive cells in the enclosed and free blades. In mice tested for contextual freezing, there was a correlation between Arc-positive cells in the enclosed and free blades. In addition, in mice irradiated with 0.5 Gy, contextual freezing in the absence of aversive stimuli correlated with the percentage of Arc-positive cells in the enclosed blade. In mice tested for cued freezing, there was no correlation between Arc-positive cells in the enclosed and free blades. In contrast, cued freezing in the presence or absence of aversive stimuli correlated with Arc-positive cells in the free blade. These data indicate that there are blade dependent effects after particulate irradiation. More studies are required to address the physiologic significance of such findings.

One of the microenvironmental factors that may influence how irradiation affects cognition is inflammation. In the current funding year we addressed this issue by irradiating animals deficient in a specific chemokine receptor: chemokine receptor-2 (CCR2). CCR2 mediates the recruitment of infiltrating and resident microglia to sites of CNS inflammation, and is upregulated by 56Fe irradiation. CCR2 knock out (KO) and wild-type mice were used to compare effects of 56Fe radiation (600 MeV, 0, 0.25 Gy) on hippocampal function using contextual fear conditioning involving tone shock pairing during training (+/+) and exposure to the same environment without tone shock pairings (-/-). In the -/- condition, irradiation increased freezing and hippocampal Arc protein levels in WT but not CCR2 KO mice, suggesting that a lack of CCR2 was associated with a poorer outcome. In contrast, in the +/+ condition, irradiation reduced freezing but there was no genotype difference. There were no significant correlations between the number of Arc-positive cells in the dentate gyrus and freezing in either genotype but in CCR2 KO mice, there was a trend towards a negative correlation between freezing and Arc-positive cells in the enclosed and free blades. In the +/+ condition, BrdU alone and BrdU/NeuN were reduced in the free blade but not the enclosed blade of CCR2 KO. CCR2 status had no influence on total numbers of newly born activated microglia before or after irradiation, indicating that other mechanisms are involved in the genotype-dependent radiation response.

Our standard markers of neurogenesis, microglia, and the immediate early gene Arc have been shown to be good predictors of radiation response after relatively high doses of photons or heavy ions. However, in our current studies they were not particularly responsive to low dose irradiation with iron ions. Given those findings we wanted to look for other markers that might track with cognition and provide predictive value after low dose exposures. The formation of long-term memory relies on modulation of synaptic connections in response to neuronal input. This plasticity requires coordinated activity-dependent synthesis of specific mRNAs and proteins that facilitate molecular and structural changes at the synapse. Dendritic spines are bulbous membrane projections of mixed morphology, that form the postsynaptic specializations of the vast majority of excitatory synapses in the central nervous system (CNS) and their structure and density are important factors in synaptic function. Spine morphology can predict both spine stability and synaptic strength, and changes in dendritic spine density or structural reorganization of spines is thought to be important for cognitive processes such as learning and memory. With the exception of our own photon data that were just recently

Task Progress:

published, little is known about the potential effects of brain irradiation on dendritic spines in the hippocampus in young adult animals.

In our new study of dendritic spines, we irradiated young adult male C57BL6 mice with 0 or 0.5 Gy (whole body) of 56Fe ions. Within each radiation treatment group, animals were randomly assigned into cohorts that received cognitive testing using a no shock/no tone paradigm or no cognitive testing, i.e. caged controls. Half of the mice from each specific treatment group were used for the determination of dendritic spine density/morphology and the other half were used for measurement of Arc and other cellular markers. Dendritic spines were visualized using a well-established Golgi staining paradigm. In the CA1 and dentate gyrus of sham-irradiated mice, the behavioral testing enhanced the spine density as compared to behaviorally naïve mice. In contrast, behavioral testing did not affect spine density in irradiated mice. These intriguing data indicate that irradiated mice did not benefit from the environmental enrichment (i.e. behavioral testing) as sham-irradiated mice did.

Finally, we initiated and completed a mixed beam study to determine how the brain responds when it is confronted by two different types of irradiation. Mice either received a single whole body exposure of 1.0 Gy protons (n = 24), a single whole body dose of 0.05 Gy 56Fe (n = 24), a single whole body exposure of 1.0 Gy protons immediately before a single whole body dose of 0.5 Gy 56Fe (n = 24) or no irradiation (n =24). Two months following irradiation, all animals receives a daily i.p. injection of bromodeoxyuridine (100 mg/kg) for 7 days, and 1 month later mice were either cognitively tested for novel object recognition or acted as caged controls. Thirty min after the onset of the last session of the cognitive testing, animals were sacrificed and their brains bisected into the two hemispheres and then frozen. Tissues were either analyzed for cellular markers of neurogenesis, the immediate early gene Arc, or for molecular/biochemical assays for markers of inflammation/oxidative stress. No cellular, molecular, or biochemical changes were observed after either of the single doses or the combined treatment. This suggests either that the low doses used did not induce any significant alterations, or the specific endpoints chosen for study were not sensitive to minor changes induced by the relatively low doses used.

Our data highlight the complexity of assessing how low doses of 56Fe affect hippocampal function. Those data suggest that even a relatively low dose of 56Fe can induce behavioral changes that involve the hippocampus, but the cellular/molecular/biochemical mechanisms responsible are still not yet known.

REPORTING FOR ANNUAL PROGRESS FROM OCTOBER 2011-JUNE 2012:

The overall goal of this project was to determine how low doses of 56Fe ions affected behavioral performance and specific cellular and molecular factors that might play a role in cognition, and if those endpoints were affected by inflammatory changes. The experimental methods chosen to accomplish these goals provided a means by which the effects of low radiation doses could be quantified both in terms of behavioral performance as well as specific cellular and molecular endpoints that are associated with learning and memory. In the current funding period we completed behavioral analyses, and analyses of neurogenesis, Arc protein expression as well as measures of the endogenous inflammatory cells in the brain, the microglia. Briefly, animals received whole body irradiation with either 0 or 0.25 Gy of 56Fe, and underwent cognitive testing 3 months later using a fear conditioning training- and testing-paradigm. The three-month time point was selected because that time has been previously used to show cognitive deficits after photon or HZE irradiation, as well as changes in Arc expression. Animals were group housed until 60 days after irradiation when they were singly housed for the duration of the experiment. Two types of behavioral paradigms were used. The first involved no tone shock pairing (-/-) and represents the ability of an animal to recognize an environment previously encountered, which requires the hippocampus. Animals that recognize an environment will explore it less, which is referred to as spatial habituation learning. The second involved tone shock pairing (+/+) and represented a Pavlovian type of response; that is, animals received a mild shock the first day, and when subsequently placed the same environment, we could determine if they remembered the adverse stimulus by showing increased freezing. Thirty minutes after the end of the final testing session, mice were sacrificed and the brains were prepared for microscopic analyses.

Because the -/- groups did not receive tone-shock pairings, we analyzed distance moved in these groups on both days. When distance moved was analyzed, there was a trend towards an effect of irradiation on the first day ($F = 3.141$, $p = 0.089$) but no effect of genotype or genotype x irradiation interaction. This trend was clearly driven by the CCR2 KO mice because those mice moved less following irradiation than sham-irradiation. This suggested either a reduced drive to explore a novel environment or that sensorimotor function might have been affected by radiation. In contrast to day 1, there was no trend towards an effect of irradiation on distance moved on day day 2, so it is more likely that the radiation effect seen in CCR2 KO mice on day 1 was due to reduced drive to explore than potential alterations in sensorimotor function. Next we analyzed freezing levels in all groups. In the -/- situation, there was a trend towards and effect of irradiation ($F = 3.401$, $p = 0.078$) and a trend towards an effect of genotype ($F = 3.498$, $p = 0.074$). WT mice that received no irradiation showed less freezing and more exploratory activity than the irradiated WT mice, suggesting that the irradiated animals showed more habituation to the environment than sham-irradiated controls. This suggested that the animals irradiated with a relatively low dose of 0.25 Gy had a better memory of the environment that sham-treated controls.

The behaviorally induced immediate early gene Arc is the most responsive immediate early gene to differences in behavioral task demands, and is essential for learning and memory. We measured Arc protein expression in terms of immunohistochemistry and western blot analysis and compared the results with our behavioral analyses. Given the well-known relationship between the various sub-regions of the hippocampus (dentate gyrus and CA areas), and because all those regions express Arc, we did a global analysis of the entire hippocampal formation using western blot technology. The results from that analysis showed an association between Arc protein and our cognitive measures, particularly in the no tone no shock condition (-/-). Immunohistochemical analyses of numbers of neurons expressing Arc protein showed a good correlation between the two blades of the dentate gyrus, a result, to our knowledge that has never before been shown. But there were no significant correlations between the numbers of Arc-positive cells in the dentate gyrus and freezing in our cognitive testing. However, there were trends toward such correlation in the CCR2 KO mice suggesting that in those animals Arc was associated with reduced associative memory formation and that this effect was dependent upon the specific region within the dentate gyrus (i.e. blade).

Measures of neurogenesis were performed in WT and CCR2 KO mice that received either 0 or 0.25 Gy and were behaviorally tested either using the -/- or +/+ tone shock pairing paradigms. BrdU alone was used as a measure of newly born cells independent of phenotype, while BrdU/NeuN specifically labeled newly born cells that differentiated into neurons. We assessed the two blades of the dentate gyrus independently. In sham animals that received -/- fear

	conditioning there were only subtle differences between genotype and morphologic area. Additionally, there were no systematic trends after irradiation, with minor decreases in cell numbers in some cases and increases in others. In animals that received +/+ tone shock pairing there was a major drop in BrdU alone and BrdU/NeuN in the free blade but not the enclosed blade of CCR2 KO. Together with the Arc data described above, these data indicate that these biomarkers of the radiation response are critically modulated by CCR2. The potential significance of those findings is not clear at this time. Overall the BrdU and BrdU/NeuN results indicate that the relationship between neurogenesis, Arc, and behavioral performance is more complex than initially anticipated. Additional studies were initiated in fund year 2012 and are continuing at the time of this writing. These include a study of dendritic spine density and morphology and a study of mixed bean irradiation. The dendritic spine study involves 48 mice and doses of 0 and 0.5 Gy of 56Fe. The mixed beam study involves 96 animals and doses of 1.0 Gy protons, 0.5 Gy 56Fe and a combination of both. Our data, while not yet complete, highlight the complexity of assessing how low doses of 56Fe affect hippocampal function, and if those effects are impacted by an altered immune function-related microenvironment (e.g. CCR2 deficiency). While the mechanisms responsible are still not yet known, our data suggest that even a relatively low dose of 56Fe can induce behavioral changes that involve the hippocampus. If or how those changes involve the process of neurogenesis, the expression of the behaviorally induced immediate early gene Arc, and the presence or absence of neuroinflammation will continue to be investigated.
Bibliography Type:	Description: (Last Updated: 04/11/2018)
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